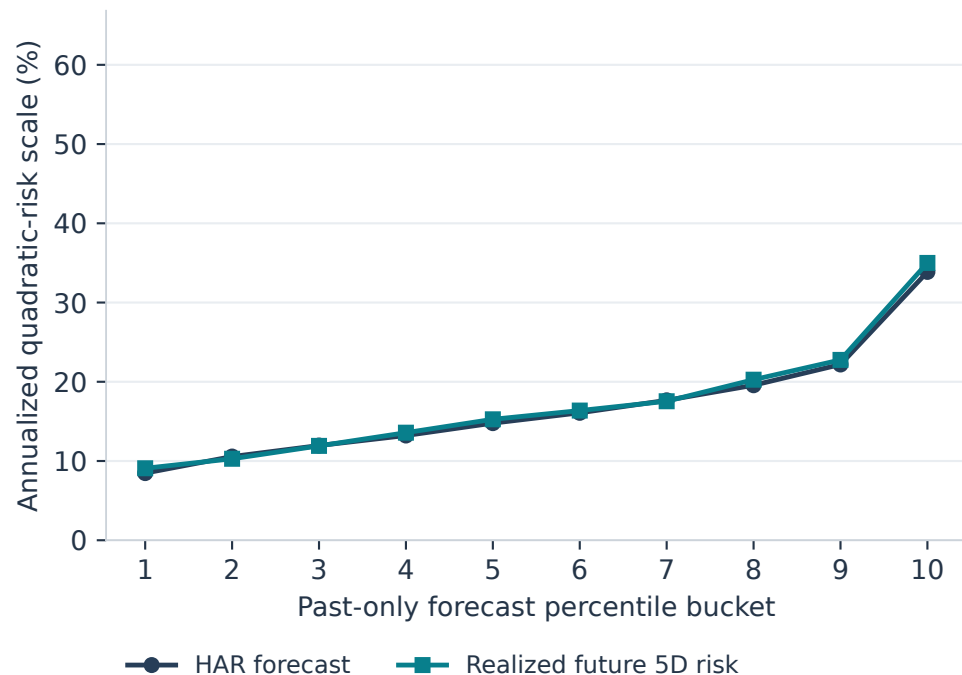


Higher forecast risk identifies higher future realized risk

Existing past-only percentile buckets | no new grouping or smoothing | EXPOSED_HISTORY

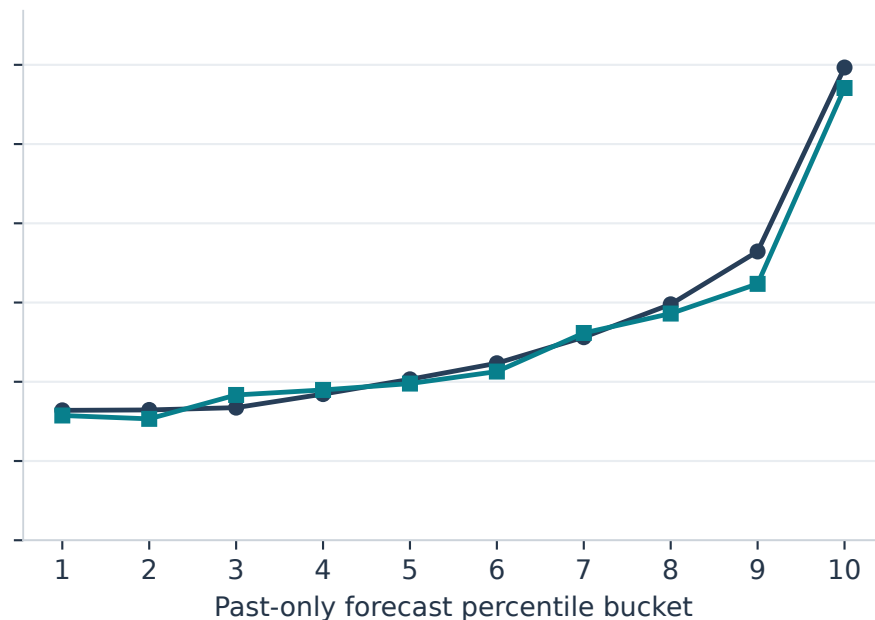
SPY | 1995-04-03 to 2026-08-20

7,646 eligible origins; bucket counts differ



QQQ | 2001-05-01 to 2026-08-20

6,112 eligible origins; bucket counts differ



Scale = $100 \times \sqrt{252 \times \text{bucket mean daily variance proxy}}$; not mean volatility or 5D endpoint-return volatility.
Buckets 1-10 span successive past-forecast percentile bands. Aggregated means describe risk ranking, not conditional coverage.